

CO3103 – PROGRAMMING INTEGRATION PROJECT
Software Engineering Track – Semester 261

DESIGN AND DEVELOPMENT OF AN ONLINE LEARNING PLATFORM

- Format:** Team-based, 6–7 students per team
- Deliverables:** Web application, source code, documentation, testing and a deployed build
- Language:** English for the reports, the presentation and the demo
- Start:** Week 38
- End:** Week 50

Project brief and assessment rubric
Semester 261

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1. Background and Objectives

Each team designs and develops an online learning platform that serves a specific group of users or a specific training context — for example university study, vocational skills, programming practice, foreign languages, or in-house corporate training.

The project pursues the following objectives:

- apply the software development process end to end: requirements elicitation, design, implementation, testing and deployment;
- build a multi-role web system with complete data and business flows;
- address a concrete learning need rather than merely cloning the interface of an existing platform;
- demonstrate technical capability through at least one advanced feature that can be measured and shown live;
- practise teamwork, source-code management, progress tracking and technical writing.

2. Representative Reference Platforms

The examples below are provided as references for product models and business flows. Teams must not copy a platform wholesale; they must identify their target users and their product's point of difference.

Platform	Aspects worth referencing
Coursera	A multi-disciplinary course catalogue, learning paths, practical assignments/projects and certificates.
edX	Standalone courses, multi-course programmes, self-paced study, graded assessments and completion certificates.
Khan Academy	Mastery-based learning, skill tracking and support for personalising the learning process.
Moodle	An open-source LMS: course and activity administration, role-based permissions, and customisation and extension through plugins.

3. User Roles

The system must support at least three roles:

1. **Learner:** search for and enrol in courses, study the material, complete assignments, track progress and interact.
2. **Instructor:** create and manage courses, content, assignments and assessments, and monitor learners.
3. **Administrator:** manage users, courses, catalogues, reports and reported content.

Teams may add roles such as teaching assistant, moderator, training manager or parent where these suit the problem.

4. Mandatory Core Functional Scope

4.1. Accounts and Authorisation

- registration, login, logout and password recovery;
- user profiles;
- role-based authorisation and API protection;
- account status management.

4.2. Courses and Content

- a course catalogue with search, filtering and pagination;
- course detail pages covering the syllabus, the instructor and the prerequisites;
- course structure organised into chapters/lessons;
- support for at least two types of learning material — for example video, text, slides or attachments;
- instructors can create, edit, publish and hide courses and content.

4.3. Enrolment and Learning

- learners enrol in or leave a course;
- view content in order and mark items as complete;
- store and display learning progress;
- resume study from the most recent position;
- display courses in progress and courses completed.

4.4. Assignments and Assessment

- quizzes and/or file-submission assignments;
- time limits or submission deadlines;
- automatic grading for quizzes;
- instructors can view submissions, grade them and give feedback;
- learners can view results and their attempt history.

4.5. Interaction and Notifications

- one interaction channel, such as per-lesson discussion, Q&A or comments;
- notifications for new content, upcoming deadlines or assessment results;
- a mechanism for reporting inappropriate content.

4.6. Administration and Basic Reporting

- user and course management;
- basic metrics on enrolments, completion rates and learning outcomes;
- an audit log of significant administrative actions.

5. Mandatory Advanced Component

Each team must select and implement at least one of the seven directions below. A team may propose a different direction if the lecturer approves it in advance.

A component counts as “advanced” only when it is integrated into the main usage flow, has supporting data or test scenarios, has evaluation criteria, and can be demonstrated live. A static interface or a single isolated API call does not meet this requirement.

5.1. Direction 1 – Grounded AI Teaching Assistant

- a Q&A chatbot grounded in the course material using RAG or an equivalent technique;
- answers must cite the relevant lesson or source;
- handle out-of-scope questions, capture user feedback, and evaluate answer quality on a test set of questions.

5.2. Direction 2 – Personalised Learning Paths and Recommendations

- recommend courses or lessons based on the learner's profile, goals, history and results;
- give a brief explanation of why each recommendation was made;
- compare against a baseline — such as most-popular or a hand-written rule — using an appropriate metric.

5.3. Direction 3 – Mastery-Based Adaptive Learning

- model skills and prerequisite knowledge;
- automatically adjust difficulty or the next lesson according to the learner's results;
- display a mastery map and demonstrate that the learning flow changes across at least two learner profiles.

5.4. Direction 4 – Automated Assessment and Academic Integrity

- a question bank that generates varied papers, automatic grading, or a secure execution environment for programming exercises;
- support for rubrics, detection of unusual similarity, or examination-session controls;
- the report must state the limitations and how mis-grading by the system is handled.

5.5. Direction 5 – Learning Analytics and Early Warning

- a dashboard analysing learning behaviour and outcomes over time;
- detect or predict learners at risk of falling behind;
- provide explanations and intervention actions; evaluate on simulated data or on data you are permitted to use.

5.6. Direction 6 – Real-Time Collaborative Classroom

- an online classroom, shared whiteboard, collaborative notes, breakout rooms or pair learning;
- state synchronisation over WebSocket or an equivalent technology;
- handle disconnection, rejoining and update conflicts to an appropriate degree.

5.7. Direction 7 – Learning Anywhere: PWA, Offline and Accessibility

- installable as a Progressive Web App, storing lessons offline and synchronising when connectivity returns;
- handle data conflicts or an offline operation queue;
- support keyboard navigation, screen readers, captions or the relevant WCAG criteria, with accessibility testing.

6. Quality Requirements

- a responsive interface, usable on both desktop and mobile;
- a clearly designed API and database, with input validation and consistent error handling;
- passwords must be hashed; sensitive information and secrets must never be committed to the repository;
- automated tests for the important business logic, and end-to-end tests for at least one main flow;
- sample data so that the lecturer can run and grade the product;
- a Git repository with branches or pull requests, issues, and a contribution history for each member;
- the product deployed to an environment the lecturer can access;
- compliance with privacy and content copyright, and clear disclosure of any use of AI or third-party services.

7. Suggested Architecture

Teams choose their own technologies but must justify the choice. A reference architecture:

- **frontend:** React, Vue, Angular or an equivalent framework;
- **backend:** Node.js/NestJS, Java/Spring Boot, Python/Django/FastAPI, .NET or equivalent;
- **database:** PostgreSQL, MySQL or another suitable DBMS;
- **storage:** object storage for video and files;
- **infrastructure:** Docker, CI/CD, a cloud deployment service;
- **observability:** logging, error tracking and a few basic operational metrics.

Using this exact stack is not required. Teams must ensure the system can be reinstalled from the documentation and does not depend on personal accounts the lecturer cannot access.

8. Milestones and Deliverables

8.1. Official Assessment Schedule

This document was updated on 3 September 2026, in week 36. Teams follow the milestones below.

Item	Deadline	Main-content limit	Weight
Proposal	23:59 Sun 20 September 2026 (end of week 38)	Max 4 pages	20%
Interim report	23:59 Sun 1 November 2026 (end of week 44)	Max 10 pages	30%
Final report	23:59 Sun 13 December 2026 (end of week 50)	Not specified	25%
Presentation and demo	Demo package ready by 23:59 Sun 13 December 2026; presentation schedule announced later	Not applicable	25%
Total			100%

The final report and demo together account for 50%: 25% for the final report and 25% for the presentation/demo.

References and Appendices do not count towards the page limit and are not themselves limited in length. However, the lecturer is not obliged to read them when grading. Students must present all important information concisely within the permitted main-content pages.

The cover page, the table of contents and all main content — including tables and figures placed in the body — do count towards the page limit. Content needed to satisfy the rubric must appear in the body; References and Appendices must not be used to evade the page limit. Marks are decided primarily on the main content and on evidence the lecturer can verify directly.

Students must read this document in full and follow it exactly. Submitting in the wrong format or the wrong language, exceeding the page limit, omitting a mandatory component, using incorrect link/access settings, or failing to meet any other requirement will incur a mark deduction.

8.1.1. Language

- **English is mandatory.** The proposal, interim report, final report, slides and all documents used for grading must be written in English.
- The presentation, the live demo and the Q&A must be conducted in English.
- The interface and the data used in the demo scenario must be consistent and clear enough to be presented in English.

8.1.2. Submission Cut-Off and Late Policy

- “End of week” means 23:59 on the Sunday of that week, Vietnam time (UTC+7).
- Any submission after 23:59 on Sunday is treated as late.
- **Each day late incurs a deduction of 20% of that item’s maximum mark.** The mark after deduction is never below 0.
- Example: the proposal is worth a maximum of 20 marks. One day late deducts 4 marks; two days late deducts 8 marks; five or more days late means the proposal scores 0.
- Teams must verify their files and links before the deadline. A corrupt file, a link without access permission, or an unreachable deployment counts as an incomplete submission until it is fixed.

8.2. Proposal – 23:59 Sunday 20 September 2026, 20%

The proposal must be a standalone document of no more than 4 main-content pages, covering:

1. the problem, the context, the target users and the product’s value;
2. a brief analysis of related solutions and the intended point of difference;
3. the functional scope, the important user stories/use cases and the acceptance criteria;
4. the chosen advanced component, how it will be integrated, and the evaluation plan;
5. the intended architecture/technology stack, and the data model or a high-level diagram;
6. a plan covering weeks 39 to 50, the assignment of work to members, and the main risks.

8.3. Interim Report – 23:59 Sunday 1 November 2026, 30%

The interim report must be no more than 10 main-content pages, covering:

1. the agreed scope and any changes relative to the proposal;
2. the UI/UX design, the architecture, the data model and the main APIs;
3. completed features, evidence that they run, and the degree of integration;
4. the prototype or initial results of the advanced component;

5. testing, security, trial deployment and technical issues;
6. progress against plan, each member's contribution, risks and the remaining plan.

8.4. Final Report and Demo – 23:59 Sunday 13 December 2026, 50%

By 23:59 on Sunday 13 December 2026, each team must submit the final report and have every resource in place for the system to be presented and demonstrated. The specific presentation and live-demo schedule will be announced later.

The final hand-over package comprises:

1. the final report, describing the final requirements, design, implementation, testing and results;
2. the source-code repository with installation/run instructions;
3. the deployed web application, with trial accounts for each role;
4. sample data, the API specification and the architecture/data-model documentation;
5. the test plan, test results and evidence of automated testing;
6. quantitative/qualitative evidence for the advanced component;
7. the slides and the demo script; a backup video is encouraged;
8. the weekly progress Google Doc, links to issues/pull requests/commits, and evidence of each member's contribution.

9. Weekly Plan

Week	Main outcome
38	Complete and submit the proposal (max 4 pages) before 23:59 Sunday 20 September 2026.
39–40	Finalise requirements, the UI/UX prototype, the architecture, the data model and the project foundation.
41–43	Implement and integrate the core features; build the advanced-component prototype.
44	Verify the integrated build and submit the interim report (max 10 pages) before 23:59 Sunday 1 November 2026.
45–46	Complete the core features and the advanced component; collect evaluation data.
47	Functional, security, usability and performance testing, and bug fixing.
48	Stabilise the deployment; finalise the evaluation, the report, the slides and the demo script.
49	Rehearse the demo, resolve any remaining defects and finalise the demo package.
50	Submit the final report and complete the product/demo before 23:59 Sunday 13 December 2026. The presentation/demo schedule will be announced later.

Teams must update the progress Google Doc every week for the whole duration of the project.

10. Team Formation and Registration

10.1. Size and Responsibilities

- each team has 6–7 students;
- each student belongs to exactly one team;
- each team elects a leader as its point of contact;
- the leader is responsible for consolidating information, sending the registration email and prompting progress updates; every member remains responsible for the accuracy of their own work.

Members of each team must make contact with one another and elect a leader as early as possible. Immediately after being elected, the leader sends the registration email and shares the progress Google Doc containing the full name, student ID and email address of every member.

10.2. Pre-Assigned Team List

The list below allocates every student on the current class rosters to a team. Teams 1–4 are unchanged from the original allocation. Teams 5–21 were drawn at random from the remaining students, with each team formed within a single class roster so that its members share a class. Every team has 6 or 7 students, apart from Team 15 (see the note below the table). Each team elects its own leader and records this in the Google Doc before the team-formation deadline.

Team	No.	Full name	Student ID	Class	Email
1	1	HÀU THÁI TÚ	2353288	A02	tu.hauthaitu@hcmut.edu.vn
1	2	Nguyễn Quang Vị	2453439	A02	vi.nguyenstudy@hcmut.edu.vn
1	3	Hàng Thái An	2452004	CC03	an.hangthaihang@hcmut.edu.vn
1	4	Võ Thanh An [†]	2452030	–	an.vo1410@hcmut.edu.vn
1	5	Kiều Gia Bảo	2410236	CC03	bao.kieugia0905@hcmut.edu.vn
1	6	Trương Lê Bảo [†]	2452147	–	bao.truong05122006@hcmut.edu.vn
2	1	TRẦN QUỐC BẢO	2352110	CC03	bao.tran2352110@hcmut.edu.vn
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12	5	Khâu Phương Nghi	2412246	L06	nghe.khaukpn@hcmut.edu.vn
12	6	Nguyễn Lê Phú Tài	2413039	L06	tai.nguyen10shi@hcmut.edu.vn
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16	2	PHAN LÊ THIÊN	2313221	L06	thien.phan2411@hcmut.edu.vn
16	3	Phạm Nguyễn Thảo Ly	2411995	L06	ly.pham241195@hcmut.edu.vn
16	4	PHẠM THÁI DUY BẢO	2352107	A02	bao.pham0716@hcmut.edu.vn
16	5	Nguyễn Ngọc Duy	2452204	A02	duy.nguyennngock24@hcmut.edu.vn

Team	No.	Full name	Student ID	Class	Email
16	6	Trần Nhân Sâm	2453120	A02	sam.transamchan@hcmut.edu.vn
17	1	Võ Phúc Khang	2452482	A02	khang.vothichanpho@hcmut.edu.vn
17	2	Nguyễn Trung Hiếu	2411002	L06	hieu.nguyen02112006@hcmut.edu.vn
17	3	Nguyễn Trường Giang	2410841	L06	giang.nguyenvt0610@hcmut.edu.vn
17	4	Trần Duy Phước Nghĩa	2412275	L06	ngheia.tranhcmut2506@hcmut.edu.vn
17	5	Trương Vỹ Kiệt	2520005	L06	kiet.truong145@hcmut.edu.vn
17	6	VŨ MINH PHÚ	2352925	A02	phu.vu150506@hcmut.edu.vn
18	1	Phan Quốc Đạt	2410717	L06	dat.phan717cs@hcmut.edu.vn
18	2	Phan Hải Phong	2412650	L06	phong.phanhai06@hcmut.edu.vn
18	3	Quách Hoàng Linh	2411895	L06	linh.quachhoang@hcmut.edu.vn
18	4	Nguyễn Thiên Phước	2453045	A02	phuoc.nguyenthien1410@hcmut.edu.vn
18	5	TRẦN MINH TÂM	2213038	L06	tam.tranminhtam@hcmut.edu.vn
18	6	Trương Hiền Minh	2452771	A02	minh.truonghien@hcmut.edu.vn
19	1	Nguyễn Thành Nam	2412187	L06	nam.nguyen48677@hcmut.edu.vn
19	2	PHẠM MINH NHẬT	2312484	L06	nhat.phamminh2312@hcmut.edu.vn
19	3	Trần Vũ Đức Hoàng	1711419	A02	hoang.tranbknetid@hcmut.edu.vn
19	4	Trần Tiến Thành	2413181	L06	thanh.tran311@hcmut.edu.vn
19	5	Võ Thái Kiệt	2411807	L06	kiet.vok24cs@hcmut.edu.vn
19	6	Võ Tấn Phát	2412616	L06	phat.votan242006@hcmut.edu.vn
20	1	NGUYỄN THỊ THÚY HẰNG	2310904	L06	hang.nguyen1809@hcmut.edu.vn
20	2	Trần Thị Yến Phương	2412794	L06	phuong.tran1284@hcmut.edu.vn
20	3	VÕ THANH NAM	2352789	A02	nam.vothanhnam22@hcmut.edu.vn
20	4	Trần Ngọc Phương Mai	2452720	A02	mai.tranngocphuongmai2452720@hcmut.edu.vn
20	5	Phạm Minh Kỳ	2411820	L06	ky.pham270106@hcmut.edu.vn
20	6	Trần Mai Hạnh Như	2452927	A02	nhu.tranakaine38@hcmut.edu.vn
21	1	Võ Nguyễn Đăng Khoa	2452571	A02	khoa.vo2452571@hcmut.edu.vn
21	2	Võ Hoàng Tuấn Kiệt	2411806	L06	kiet.vo2186@hcmut.edu.vn
21	3	TẠ NHẬT LINH	2252432	A02	linh.tanhat2004@hcmut.edu.vn
21	4	Văn Thị Như Quỳnh	2412984	L06	quynh.vannhuquynh06@hcmut.edu.vn
21	5	Nguyễn Võ Hoàng Sơn	2453128	A02	son.nguyenhoang24@hcmut.edu.vn
21	6	Trần Mậu Giàu	2410851	L06	giau.tranmau@hcmut.edu.vn

† This student appears in the original allocation but was not found on any current class roster. Their enrolment must be confirmed before the team-formation deadline.

Team 15 has only 3 students: the TN02 roster (course 148770) contains six students in total, three of whom are already in Team 4. This team requires the lecturer's approval or a correspondingly reduced project scope.

10.3. Team-Change Rules

- a student who wants to change team must take the initiative to find a student in another team who agrees to swap places directly;
- a change is considered only when both parties agree. The two students involved and both team leaders must confirm their agreement in the same email thread;
- one of the two students emails the lecturer requesting the swap, copying (CC) the other student and both team leaders. The email must state the full name, student ID, current team and requested team of both students;
- a swap must preserve the required team sizes and takes effect only after the lecturer confirms it. Do not change names in the Google Doc before receiving confirmation;
- swap requests should be sent as early as possible so as not to disrupt leader elections, work allocation and proposal preparation. All changes require the lecturer's approval.

Subject line for a swap request:

DATH-261-SWAP-<Student 1 name>-<Student 2 name>

Email template:

Dear Lecturer,
 We would like to request the following group swap:
 Student 1: <Full name>, <Student ID>, from Group <A> to Group
 Student 2: <Full name>, <Student ID>, from Group to Group <A>
 Both students agree to this swap. The leaders of Group <A> and Group are copied on this email and confirm their agreement by replying to this email thread.
 We understand that the swap is effective only after the lecturer's confirmation.
 Kind regards,
 <Full name>
 <Student ID>

10.4. Registration Email

Immediately after the team is formed, the leader emails the lecturer with the subject line:

DATH-261-<Team leader name>

For example: DATH-261-Nguyen Van An.

The email must contain at least:

- the team name;
- the full name and student ID of the team leader;
- the list of members;
- the intended context/type of online learning platform;
- the intended advanced direction;
- a link to the progress Google Doc, shared with the correct permissions.

The Google Doc is a live document that must be updated weekly. The leader should include the Google Doc link in the email or attach it via Google Drive — do not send an exported PDF/DOCX in place of the shared link. The document must grant the lecturer access, and team members must have edit permission.

10.5. Team Registration Email Template

Dear Lecturer,
 My name is <Full name of group leader>, Student ID <Student ID>, and I am the representative of <Group name>. Our group consists of <6/7> members and would like to register an online learning web application project with the following direction: <Brief description>.
 Proposed advanced component: <Name and expected outcome>.
 Weekly progress Google Doc: <Google Doc URL>
 The Google Doc contains the full name, student ID, and email address of every member. We have granted the lecturer access and all members editing permission.
 Kind regards,
 <Full name of group leader>
 <Student ID>
 <Email>
 <Group name>

10.6. Interim Report Submission Email Template

The team leader submits the interim report by email, in English. Subject line:

DATH-261-INTERIM-<Team name>-<Team leader name>

The email must attach the interim report as a PDF and provide the links the lecturer needs in order to check progress.

```
Dear Lecturer,
On behalf of <Group name>, I am submitting our interim report for
CO3103 – Programming Integration Project (Software Engineering),
Semester 261.
Group leader: <Full name>
Student ID: <Student ID>
Project title: <Project title>
Submission package:
1. Interim report (PDF, maximum 10 main-content pages): <Attached/Link>
2. Deployed prototype: <URL>
3. Source-code repository: <URL and branch/tag>
4. Weekly progress Google Doc: <URL>
5. Supporting demo or evidence, if any: <URL>
Current advanced component: <Name and one-sentence status>
Current overall status: <On schedule/At risk/Delayed, with brief reason>
We have verified that all files and links are accessible to the lecturer.
Kind regards,
<Full name of group leader>
<Student ID>
<Email>
<Group name>
```

10.7. Final Report and Demo Package Submission Email Template

The team leader submits the final report and the demo resources by email, in English. Subject line:

DATH-261-FINAL-<Team name>-<Team leader name>

```
Dear Lecturer,
On behalf of <Group name>, I am submitting our final report and
demo-ready project package for CO3103 – Programming Integration
Project (Software Engineering), Semester 261.
Group leader: <Full name>
Student ID: <Student ID>
Project title: <Project title>
Advanced component: <Advanced component name>
Final submission package:
1. Final report (PDF): <Attached/Link>
2. Production deployment: <URL>
3. Demo accounts and credentials: <Secure location/instructions>
4. Source-code repository: <URL and final release/tag/commit>
5. Installation and reproduction guide: <URL>
6. Test report and advanced-component evaluation: <URL>
7. Presentation slides: <Attached/Link>
8. Backup demo video, if available: <URL>
9. Weekly progress Google Doc: <URL>
We confirm that the submitted system and demo package are ready for
the presentation and live-demo schedule to be announced later. We have
verified that all files, accounts, and links are accessible to the lecturer.
Kind regards,
<Full name of group leader>
<Student ID>
<Email>
<Group name>
```

11. Weekly Progress Google Doc Template

Suggested document name: DATH-261-<Team name>-Weekly-progress.

11.1. Team Information

No.	Full name	Student ID	Email	Role / responsibility
1				Team leader
2				
3				
4				
5				
6				
7				Delete this row for a team of 6

11.2. Project Information

- Project title:
- Target users:
- Problem to be solved:
- MVP scope:
- Chosen advanced component:
- Source-code repository:
- Deployment:
- Task-management board:

11.3. Weekly Progress Log

Create one entry per week, containing:

1. the goals for the week;
2. work completed, per member, with issues and evidence;
3. work not completed, and why;
4. issues/risks that need support;
5. the plan for the following week;
6. significant team decisions.

Links to issues, pull requests, commits, designs or demo videos should be used as evidence. Do not simply record an overall percentage without naming concrete deliverables.

12. Assessment Rubrics

12.1. How the Rubrics Are Applied

Each criterion is graded on how fully it is met and on the actual evidence provided:

- **Excellent – 100% of the criterion’s marks:** complete, correct, consistent, with convincing evidence and clear analysis;

- **Good – approximately 75%:** most requirements met, with minor shortcomings that do not affect the main objectives;
- **Partially met – approximately 50%:** results exist but lack depth, are poorly integrated, or the evidence is weak;
- **Not met – 0–25%:** missing, incorrect, not working, unevidenced or unexplainable.

The lecturer may award marks between these levels according to actual quality. A team's mark may be adjusted for individual contribution as evidenced by the Google Doc, issues, pull requests, commits, the report and the Q&A.

12.2. Proposal Rubric – 20%

Criterion	Marks	Requirement for full marks
Problem, context and users	3	States the real need, the target users, the pain points and the product's value.
Requirements and scope	4	Features, user stories/use cases, MVP, out-of-scope items and acceptance criteria are clear and feasible.
Solution and high-level design	4	Main flows, architecture, data, UI/UX and technology choices are sound and consistent.
Advanced component	4	A specific direction chosen, with its difficulty, integration approach, data/experiments and evaluation metrics explained.
Implementation plan	3	Milestones for weeks 39–50, work allocated across all 6–7 members, dependencies, risks and mitigations.
Proposal quality	2	Clearly presented, within the 4-page limit, legible figures/tables, correct citations and no significant contradictions.
Total	20	

12.3. Interim Report Rubric – 30%

Criterion	Marks	Requirement for full marks
Progress against the proposal	5	A transparent goal-versus-result comparison, explanation of changes, and a recovery plan where behind schedule.
Requirements, UI/UX and architecture	5	Design updated to reflect development; flows, data, APIs and system components are consistent.
Core feature implementation	7	The important flows run and are integrated; evidence from the deployment, screenshots or video, and sample data.
Advanced component	5	An integrated prototype or initial results; a clear evaluation plan, data, baseline/metrics and risks.
Software engineering and quality	5	Git/issue management, testing, error handling, basic security, and appropriate CI/CD or trial deployment.
Report and team contribution	3	Clear report within the 10-page limit, with evidence and a concrete account of each member's work.
Total	30	

12.4. Final Report Rubric – 25%

Criterion	Marks	Requirement for full marks
Problem, requirements and traceability	4	Describes the final requirements and traces them through design, implementation and testing.
Architecture and implementation	5	Explains technical decisions, the data model, APIs, the main components and trade-offs, with evidence from the system.
Advanced component	5	Describes the method, integration, data/experiments, baseline or control, and a well-grounded analysis of results.
Testing and system quality	5	A test strategy, test cases/results, automated testing, security, performance/reliability and defect analysis.
Deployment and reproducibility	3	An accessible deployment; instructions, configuration, sample data and demo accounts that allow the system to be re-run.
Report quality and project reflection	3	Clear writing, correct figures/tables and citations; states limitations, lessons learned, individual contributions and future work.
Total	25	

12.5. Presentation and Demo Rubric – 25%

Criterion	Marks	Requirement for full marks
Demo of the core flows	6	A live, uninterrupted demonstration of the main learner, instructor and administrator flows using realistic data.
Demo of the advanced component	6	The advanced feature works within the product; the team demonstrates a meaningful scenario and explains the evaluation results.
Stability and user experience	4	The system responds well, the UI/UX is consistent, errors are handled sensibly, and there are no serious defects in the main scenario.
Presentation quality	3	Clear structure, within the allotted time, focused on problem–solution–evidence rather than a list of features.
Technical understanding and Q&A	4	Able to explain the architecture, the code, the data, the technical choices, the limitations and the team’s decisions.
Member participation	2	Roles sensibly divided; members understand their own work and contribution.
Total	25	

13. Minimum Acceptance Criteria

The project is considered eligible for the final demo when:

- the deployment is reachable and can be logged into with the trial accounts;
- the main flows for learner, instructor and administrator are complete;
- the advanced component works within the product, with a demo scenario and evaluation results;
- there are no serious defects causing data loss or bypassing authorisation;
- the source code, sample data, installation documentation and test report have been handed over;
- the progress Google Doc is kept up to date and reflects each member’s contribution.